Heterogeneous Binary Pixel Difference Networks for Remote Sensing Object Detection

Jialei Zhan, Liang Bai, Jiehua Zhang, Tianpeng Liu, Fan Shi, Yongxiang Liu, and Li Liu

***Abstract*— Recent research in remote sensing object detection (RSOD) has significantly advanced the development of vision foundation models. However, deploying these models on resource-constrained edge devices is challenging due to their high computational demands. Binarized detectors utilize binary neural networks (BNNs) to achieve extreme compression by quantizing weights and activations to $+1$ or -1, which have been extensively studied for generic object detection tasks. In remote sensing images, the objects of interest typically exhibit weak responses, and the images often contain numerous unique local areas. Feature binarization in these images can lead to substantial loss of object contrast and scale prior information, which exacerbates performance issues, particularly for small objects, resulting in significant performance degradation. To address these challenges, we propose a novel binarized detector for RSOD named the heterogeneous binary pixel difference network (HBiPiDiNet). Initially, we developed a binary pixel difference convolution (BiPDC) that integrates local binary patterns (LBPs) to capture local contrast information with traditional binary convolution, thereby enhancing the representation of small objects. Subsequently, we constructed heterogeneous kernel fusion convolution blocks (HKFCB) based on BiPDC and standard binary convolution. The HKFCB comprises multiple BiPDCs at different scales, effectively representing BiPDC under multiscale LBP and multiscale binary convolutions. Extensive experiments demonstrate that our proposed method significantly enhances the performance of state-of-the-art binary detection methods across three remote sensing datasets: AI-TOD, VisDrone2019, and DIOR. We have released our code and models at https://github.com/yuhua666/HBiPiDiNet/tree/main.**

***Index Terms*— Binarization, binary neural networks (BNNs), local binary patterns (LBPs), object detectors, remote sensing.**

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I. Introduction

REMOTE sensing object detection (RSOD) utilizes optical remote sensing images to detect and pinpoint ground objects like vehicles, planes, and ships [1]. Recent research in RSOD has significantly advanced the development of vision foundation models [2], [3], [4], [5]. However, the complexity and computational demands of deep neural network architectures, such as DCNNs [6] and ViTs [7] that are essential for object detection, pose significant challenges for resource-constrained devices such as drones and satellites. The high resolution of remote sensing images significantly increases the computational burden. For example, using Faster RCNN with a ResNet18 backbone on the AI-TOD dataset requires 101.42 GFLOPs, which fails to satisfy the efficiency standards required for effective object detection. Recent advancements in foundational models for remote sensing, such as RingMo [8], Plain ViT [9], Rsprompter [10], and SpectralGPT [11], have significantly increased both the size and computational requirements. Specifically, Plain ViT comprises up to 100 million parameters, whereas SpectralGPT contains 600 million parameters. Owing to the high resolution of remote sensing images, the models designed for such applications must be deployed on satellites, drones, and similar equipment in practical scenarios. The efficient design of remote sensing models aims to reduce power consumption and facilitate deployment on these devices, thereby enabling the widespread application of remote sensing models. Driven by the substantial overheads of increasingly deeper network structures and large-model technologies, there is a pressing need to develop more compact and efficient detectors for RSOD.

The objective of compact model design is to develop a more efficient network architecture, significantly reducing computational costs while maintaining accuracy. The fundamental challenge in developing compact and efficient object detectors is finding a balance between achieving high accuracy and maximizing efficiency [12]. Current RSOD methods employ a range of model compression techniques including compact structural design [13], quantization [14], distillation [15], and low-rank decomposition [16] to reduce their reliance on computational resources. Among these methods, quantization effectively reduces the bitwidth of weights and activations, thereby achieving more efficient inference [17]. As the most extreme form of quantization, binary neural networks (BNNs) constrain the weights and activations to either $+1$ or -1, thereby reducing storage and computational costs by factors of 32 and 64, respectively [18].

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Owing to their extremely high energy efficiency, BNNs have been extensively researched for application in generic object detection tasks [19], [20]. We expect that this binarization technique will inspire the development of efficient RSOD networks. Although binary detectors are more efficient than their real-valued counterparts, their capacity to represent data diminishes, causing significant feature response loss and reduced detection performance, especially for small objects in remote sensing. Conventional binary detection frameworks are primarily designed for natural images with larger and more uniformly distributed objects; however, they often fail to capture key local features and lack multiscale representation when applied to remote sensing. This frequently results in lower detection accuracy and increased false negatives (FNs) and false positives (FP) in RSOD. To overcome these challenges, this study introduced a novel binary detection framework specifically tailored for RSOD, named heterogeneous binary pixel difference network (HBiPiDiNet).

Compared to natural optical images, remote sensing optical images exhibit less pronounced responses and nonuniform object distribution. In RSOD, the feature binarization process significantly diminishes information and impairs the model to distinguish small objects from the background, thereby weakening key feature responses. This significantly compromises the performance of the existing binary detectors. This study addresses two main challenges in effectively binarized RSOD networks.

1) *Low response from object features and less contrast with surrounding objects*: Compared with natural images, remote sensing images typically occupy smaller pixel areas. In RSOD, the binarization process exacerbates the potential confusion between objects of interest and surrounding irrelevant objects, leading to higher rates of FPs and FNs [refer to Fig. 1(b)].
2) *Challenging to effectively detect nonuniform distribution of objects:* In remote sensing images, certain areas exhibit nonuniform distribution of objects, with several regions lacking any objects of interest, whereas other areas exhibit densely packed objects. The binarization operation further weakens the ability of the model to capture features of minute objects, thereby challenging the effectiveness of binarized networks in identifying areas with both dense and nonuniform object distributions.

To enhance the accuracy of object detection in remote sensing images using binary detectors, it is crucial to integrate prior knowledge into the feature binarization process, such as local contrast information. LBP is a traditional handcrafted operator that excels in capturing local contrast in images. Drawing inspiration from previous research [21], we incorporate LBP into the feature binarization representation. The incorporation of LBP, which is known for capturing local edge details, has been explored in recent studies [22], [23], where it is combined with convolutional operators to enhance the representation of high-frequency information. This enhancement improves the model's response to local features. Such an approach enables the binary detector to retain edge and local contrast details of small objects in remote sensing images, thereby

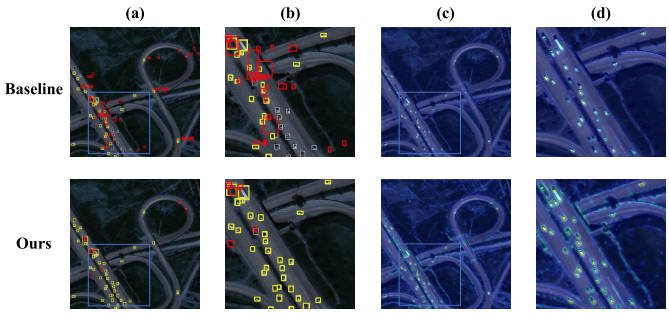


Fig. 1. Illustration of the effectiveness of our method in remote sensing scenarios. The yellow bounding boxes indicate correctly detected objects and red bounding boxes signify erroneously detected objects. White bounding boxes denote missed detections. (a) Detection results and (c) corresponding heatmaps, respectively. (b) and (d) Blue boxes area of (a) and (c).

improving its performance. As depicted in the first column of Fig. 1, the binarization of features in remote sensing images obscures critical object information and diminishes the edge response of the objects. This results in blurred boundaries between objects and the background, as well as indistinct edges of individual objects, which complicates detection tasks by increasing the likelihood of FPs and missed detections. The introduction of the LBP operator to enhance local response information of objects has significantly strengthened the edge response of object features, thereby improving the detection accuracy of small objects (refer to Fig. 1). Our method reduces both FPs and FNs, particularly by enhancing the detection of small objects in densely populated scenes. Based on heatmap analysis, our method substantially improves the feature response of the objects of interest. A weak feature response in the baseline can result in the omission of object features in the model, whereas the enhanced feature response of this method aids the model in feature extraction, leading to performance improvements in the corresponding images. In this study, we achieved both high precision and high efficiency in RSOD by employing binary networks for extreme compression, complemented by the heuristic operator LBP. Binarization results in information loss, including the loss of crucial prior information in RSOD, such as contrast and scale priors [24], [25]. To capitalize on these essential priors, we aimed to integrate the LBP operator with binary networks. This integration focuses on leveraging the contrast and scaling prior information of objects in remote sensing images to enhance the performance of binary networks. In general object detection, visual cues in natural images primarily include features such as color, texture, and shape. These cues differ markedly between object categories and between objects and their backgrounds, reducing potential confusion, as objects typically occupy larger pixel areas. However, in RSOD scenarios, these features are less pronounced owing to the smaller size of the objects. The information loss caused by neural network downsampling and binarization exacerbates this issue, leading to a significant decline in detector capabilities. When the color, texture, shape, and other intrinsic features of an object are not prominent, the network can recognize and locate the object by analyzing differences between the object and its surrounding background. This is achieved by comparing pixels or regions [26], [27] in areas adjacent to the feature space

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to capture unique and densely arranged objects in the scene. Therefore, binary pixel difference convolution (BiPDC) [21] has been integrated into binary networks to emphasize contrast priors, thereby enhancing feature responses in key areas and improving the overall performance. Building on these insights, we introduce BiPDC to address the challenges of low response from object features and low contrast with surrounding objects. This innovation combines the local binary pattern (LBP) operator with binarized convolutions, enhancing local contrast information crucial for remote sensing. This research presents a parallel form of BiPDC, which is more suitable for RSOD than the original form. The parallel BiPDC consists of two branches: BiConv, which computes pixel intensity differences, and BiPDC [21], which integrates the LBP mechanism into basic convolution operations to detect local pixel value differences. This dual approach not only preserves the ability to compute pixel intensity differences but also augments the capacity of the network to perceive contrast priors. It mitigates the risk of binary networks losing macroscopic image analysis capabilities due to diminished pixel intensity difference computation and improves the discernment of local contrast priors, thus facilitating object-background differentiation. Notably, BiPDC maintains the same computational efficiency as binary convolution, maintaining the model completely binarized. To further enhance performance and address the challenge of effectively detecting situations with uneven object distribution, we integrated heterogeneous kernel fusion convolution blocks (HKFCB) into the network architecture. By transforming homogeneous convolutions into heterogeneous ones within the feature pyramid network (FPN) and incorporating nonhomogeneous BiConv and BiPDC operations, the network becomes adept at highlighting local critical responses of objects and enhancing the detection of objects of various sizes. Additionally, the design of a heterogeneous convolutional structure further addresses the issue of nonuniform distribution of objects. The contributions of this study can be summarized as follows.

1) We have developed a new convolution operation, termed BiPDC, which is specifically designed to utilize local contrast priors. This method improves the response in crucial local regions, thereby enabling the network to capture highly discriminative information.
2) To address the challenge of irregularly shaped objects in remote sensing images, we introduce the HKFCB. This technique incorporates a heterogeneous kernel structure that combines BiPDC and BiConv with different kernel sizes tailored to various FPN configurations. The design employs larger convolution kernels at higher levels to enhance contextual information, whereas smaller kernels at lower levels minimize the risk of incorporating irrelevant information from outside small objects.
3) We conducted extensive evaluations of our proposed method on three RSOD datasets, performing a detailed comparison with leading binary detectors in the field. This approach not only achieved optimal accuracy but also maintained a computational complexity comparable to these advanced binary detectors. Additionally, experiments with lightweight RSOD methods using the extended binarization technique demonstrated the broad applicability of our method across different types of object detectors, confirming its versatility.

II. Related Work

This section provides a concise review of the recent advancements in lightweight object detection that have led to significant developments in RSOD, BNNs, binary object detectors, and traditional operators that enhance convolution.

A. Lightweight RSOD

Efficient neural network strategies mainly include model lightweight design, data refinement design, and model's efficient fusion of data and features. Model lightweight design [28], [29] is a common efficient model strategy in deep learning, aimed at achieving models with low parameters and low computational requirements while maintaining acceptable accuracy, such as pruning [30], quantization [31], and distillation [32]. Data refinement design aims to improve data efficiency and optimize data structures, such as sample pruning [33]. The model's efficient fusion of data and features [34] primarily involves learning methods using data from different modalities. For example, Hong et al. [35] have studied the intrinsic impact of large distribution gaps between HS-MS data by combining spatial and spectral feature differences, proposing a sub-pixel level HS super-resolution framework through designing a new decoupling-coupling network. Moreover, for multicity remote sensing scenarios, Hong et al. [36] have established a new multimodal remote sensing benchmark dataset and proposed a high-resolution domain adaptive network to enhance the model's generalization ability in multicity environments. Based on above discussion, this article focuses on developing practical lightweight RSOD models for single remote sensing scenarios, emphasizing model efficiency and thus concentrating on research to enhance model lightweight effectiveness.

Deep learning-based RSOD models have become increasingly popular [37], but challenges persist, especially with detecting small and variably scaled objects and managing limited feature information [38]. Current strategies primarily aim to enhance the representation of small objects through methods such as feature imitation [39], [40], data augmentation [41], [42], multiscale learning [43], [44], and advanced training strategies [45], [46]. Despite these efforts, the performance of these methods still heavily relies on computationally demanding detectors, which complicates the deployment on resource-constrained devices. To address the need for both high accuracy and efficiency, various model compression techniques have been explored. These include efficient network design [13], knowledge distillation [15], [47], [48], network pruning [49], [50], and model quantization [14]. For instance, Pang et al. [13] introduced a lightweight residual backbone that optimizes computation and memory usage. Yang et al. [47] developed an adaptive reinforcement-supervised distillation framework to improve the capability of lightweight models in detecting larger objects through distillation. Li et al. [15] leveraged instance-related information to enhance knowledge transfer in teacher–student model frameworks by integrating remote sensing scene features with instance-level relationships.

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Chen et al. [49] designed a backbone structure and applied network pruning to achieve channel-level sparsity, thus creating compact models that maintain efficiency in ship detection. They also employed knowledge distillation to mitigate performance losses due to network pruning. Shen et al. [50] proposed a multivolume YOLOv4 model that achieves a balance between efficiency and accuracy in RSOD using a network pruning strategy. Zhang et al. [48] introduced a dynamic global distillation module that identifies valuable regions in both foreground and background for multiscale feature imitation. Furthermore, Zhang et al. [14] proposed a data-free quantization method to reduce the bit width of detectors and accelerate RSOD inference. This study focuses on BNNs, which represent an extreme form of model quantization.

B. Binary Neural Network

The concept of neural network binarization was introduced in EBP [51] and BNN [52], which developed a comprehensive framework for training networks with binary weights and activations using gradient backpropagation. Despite this advancement, the accuracy of such networks on large-scale datasets like ImageNet remains relatively low. Subsequent studies have focused on enhancing this precision. For instance, XNOR-Net [53] improved binarization by multiplying binary weight kernels with real-valued scale factors, significantly narrowing the performance gap with real-valued ResNet-18. Building upon XNOR-Net, Bi-RealNet [54] introduced shortcuts that propagate real values across feature maps, thereby increasing the accuracy of binarized networks on extensive image-classification datasets. Furthermore, because of the performance sensitivity of binary networks to variations in activation distribution, ReactNet [55] proposed generalized versions of the traditional Sign and PReLU functions, named RSign and RPReLU, respectively. These functions are designed to learn distribution reshaping and offsetting actively, using distribution loss to encourage binary networks to emulate the output distribution of real-valued networks. The rigidity of BNNs leads to a degradation in the quality of binary activations and weights. To address this, INSTA-BNN [56] dynamically adjusts quantization thresholds in a manner dependent on the input or instance, using higher-order statistical moments to accurately estimate the input distribution characteristics. Large pre-trained vision transformers (ViTs) [57] have demonstrated impressive performance across various visual tasks. However, their deployment on resource-limited devices is hindered by high computational and memory demands. To mitigate these challenges, researchers have begun exploring binarization techniques specifically for transformers. Initial efforts in the NLP field led to the development of BiT [58]—a fully binarized transformer model featuring an elastic binary activation function with learnable parameters and a method for extreme network quantization through continuous distillation of high-precision models into lower-precision students. In the field of computer vision, Q-ViT [59] has investigated binarization strategies for ViTs, introducing an information rectification module and a distribution-guided distillation technique to fully quantize ViTs, thereby eliminating distortions and achieving precise, low-cost implementations. Additionally, BinaryViT [60] has merged key architectural elements of binarized CNNs with a pure ViT structure, thereby enhancing the representational power of binary ViTs without incorporating convolutions, and surpassing the performance of state-of-the-art binary CNNs. Given the challenges posed by the long-tailed distribution of Softmax attention in ViT structures, which significantly impacts binarization accuracy, BiViT [61] introduced a Softmax-aware binarization approach. This method dynamically adapts to the data distributions and reduces binarization errors. It also employs a cross-layer binarization strategy that separates the binarization processes of self-attention and multilayer perceptrons (MLPs), incorporates parameterized weight scale factors for weight binarization, and effectively reduces errors associated with binarization. In summary, both binary CNNs and binary ViTs have undergone considerable advancements. Building on these developments, we propose a binary remote sensing object detector that integrates both CNN and ViT backbones.

C. Binary Object Detectors

Owing to the high efficiency of BNNs, numerous binary detectors have been developed for generic object detection tasks. These detectors can also enhance the performance of RSOD. Among these methods, BiDet employs the information bottleneck (IB) principle to eliminate redundancy and effectively utilize the representational capacity of binary convolution. This method represents the first binarized network designed specifically for efficient object detection. ASAD-FRCNN [62] introduced simple and efficient loss functions that suppress the shared magnitude between real and binary kernels, enabling significant model compression in the binary Faster-RCNN detector. LWS-Det introduces an automated layer-by-layer search strategy with angle and magnitude loss functions to increase the capacity of binary detectors and enhance their performance to approach that of real-value networks. Similarly, IDa-Det [20] utilizes the Mahalanobis distance to measure the information discrepancy between the proposals of teacher and student models, effectively reducing the performance gap between binary and real-valued detectors. Despite these advancements, the presence of small and diverse object scales in remote sensing scenes has posed challenges in developing effective binary detectors for RSOD.

D. Traditional Operators Enhance Convolution

The LBP [63] is a visual descriptor used in computer vision classification. Recent studies have integrated LBP with convolution to reduce network complexity and improve feature representation by computing pixel differences during the convolution process. Local binary convolution (LBC) [64] was inspired by LBP as an efficient alternative to standard convolutional layers using a set of fixed sparse predefined binary convolution filters that are not updated during training, a nonlinear activation function, and a set of linear weights that are learnable. Similarly, central differential convolution (CDC) [22] employs learnable weights to capture the image gradient information by aggregating the grayscale and gradient information and capturing the intrinsic detailed patterns of an image for facial anti-spoofing tasks. Pixel difference network

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(PiDiNet) [23] proposed a new pixel difference convolution (PDC) that integrates traditional edge detection operators such as Sobel, Candy, and LBP into popular convolution operations in modern CNNs. In addition to CPDC based on LBP and benefiting from ELBP [65], PiDiNet introduced APDC and RPDC based on AD-LBP and RD-LBP, respectively. It compares neighboring pixels on circular and radial grids to detect intensity variation patterns in the angular and radial directions, encoding the relationships between pixels inside the convolution kernel. This allows it to incorporate both the high-level semantic information-processing capability of the convolution and the local detail-capturing ability of traditional operators. Building on this, Su et al. [21] extended PDC to its binarized form, BiPDC, to address the inherent issue in prior binarization networks. They utilized a shared threshold for the entire image during the binarization process and disregarded the local content variations, which caused an irreversible loss of high-order image details. BiPDC employs dynamic thresholds based on neighboring pixels for binarization, thereby enabling the preservation of high-order texture details within the binarization network. Building on previous research, variations of BiPDC have been developed. Considering that noise-corrupted image patterns can lead to incorrect activations, thereby significantly reducing the accuracy of face recognition in noisy environments, and recognizing the inherent robustness of CNNs to various noise levels, Zhang et al. [66] proposed a median pixel difference convolutional network (MeDiNet) [66]. In this model, traditional convolutional layers are partially replaced with median PDC layers, integrating the concept of traditional multiscale median filtering with deep CNNs. This modification effectively eliminates the noise pixels in the feature maps and mitigates the adverse effects of noise. In our study, we have expanded BiPDC into a parallel form, which will be elaborated in the Methods section. Unlike BiPiDiNet, our approach concentrates on RSOD. We incorporated a parallel binary convolution to enhance macrosemantic information, ensuring a balance between macrosemantic information and local contrast, thus improving the detection capabilities of the model.

III. Baseline BNN of RSOD: Binary CNNs and ViTs

As the field of binarization networks progresses, existing models have shown substantial improvements in accuracy, achieving impressive results in tasks such as image classification and object detection. However, because of the challenges of weak object responses and varying scales in remote sensing scenarios, no binarization networks currently cater to RSOD. Therefore, this study aims to develop a binarized object detector suitable for remote sensing scenes, drawing inspiration from state-of-the-art (SOTA) binarization methods used in natural image object detection. To this end, a baseline model was constructed based on these existing methods. In our research, we adopted the state-of-the-art IDa-Det as the detection framework [20]. We excluded the knowledge distillation steps designed for natural images in this network. For quantization of the CNN and ViT backbones, ReactNet [55] and BiT [58] were utilized. Additionally, RFLA was selected as the label allocation strategy appropriate for remote sensing scenarios [67], and its details are discussed in this section.

A. Implementation of Binary Convolutions

In a specific convolution layer, $w \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K \times K}$, $x_{\text{in}} \in \mathbb{R}^{C_{\text{in}} \times W_{\text{in}} \times H_{\text{out}}}$, and $x_{\text{out}} \in \mathbb{R}^{C_{\text{out}} \times W_{\text{out}} \times H_{\text{out}}}$ denote kernel weights and feature maps, where C_{in} and C_{out} denote the number of input and output channels, H and W represent the size of feature maps, and K denotes the kernel size. The convolution operation can be formulated as

$$x_{\text{out}} = x_{\text{in}} \otimes w \tag{1}$$

where $\otimes$ denotes a convolution operation. To simplify the notation, we omitted the batch normalization layer and activation function. Based on (2), the binarized convolution operation can be formulated as follows:

$$x_{\text{out}} = a \circ x_{\text{in}}^{b} \odot w^{b} \tag{2}$$

where a denotes the channel-wise scale factor used to minimize quantization error, $\odot$ denotes the XNOR and bit count operations, w^b and x_{in}^b denote the quantization of w and x_{in} into $\{+1, -1\}$ using the Sign function. The Sign function is defined as

$$\text{Sign}(x) = \begin{cases} +1, & x > 0 \\ -1, & x \leq 0. \end{cases} \tag{3}$$

Herein, following the methodologies in [20] and [55], a channel-wise learnable parameter is added before the Sign function when processing feature maps to effectively enhance the representation capability of BNNs.

B. Implementation of Binary MSA and MLP of ViT

Previous research on BNNs has predominantly focused on binarizing the ResNet architecture. Given that the ViT demonstrates outstanding performance but suffers from high computational and storage demands, we selected Poolformer and PVT to develop a binary ViT backbone. As described in [57], the basic transformer block in ViT consists of a multihead self-attention (MSA) layer and an MLP. The transformer block is fully binarized using a binary function as per BiT [58]. The weights of the fully connected layers in the transformer blocks are binarized to $\{+1, -1\}$. The activations are binarized to $\{+1, 0\}$ for Softmax outputs and $\{+1, -1\}$ for other layers. For the binarization of weights w in fully connected layers, zero-mean and scale factors are applied to enhance the information capacity, which can be described as follows:

$$\text{Bi-linear}(x_{\text{in}}) = a_w \left(x_{\text{in}}^{b} \odot \text{Sign}(w - \mu(w)) \right) \tag{4}$$

where w denotes the weight of the fully connected layers, $\mu(\cdot)$ indicates the mean value, and a_w denotes the weight scaling factor. In the attention layer, we employed the elastic binarization function introduced by BiT [58] for the binarization of the activations. The calculation can be described as follows:

$$\text{Bin-att}(x_{\text{in}}) = \alpha \left\lfloor \text{Clip}\left(\frac{x_{\text{in}} - \beta}{\alpha}, 0, 1 \right) \right\rceil \tag{5}$$

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where α and β are learnable scale factor and the threshold, respectively. α is initialized as $(\|x_{\text{in}} \cdot \mathbf{1}_{\{x_{\text{in}} \geqslant 0.5\}}\|_{l1}/n_{\{x_{\text{in}} \geqslant 0.5\}})$ and β is initialized as 0.

For the other activation binarization, the calculation can be described as

$$\text{Bin-act}(x_{\text{in}}) = a \cdot \text{Sign}(x_{\text{in}}), \quad a = \frac{1}{n}\|\mathbf{x}_{\text{in}}\|_{l1}. \tag{6}$$

Based on (4)–(6), the computation process of the binarized transformer is outlined. Within a specific transformer layer, the attention head computation is described as follows:

$$Q, K, V = \text{Bi-linear}_{Q,K,V}\left(x_{\text{in}}^b\right) \tag{7}$$

where x_{in}^b represents the binarized output of the input, and $\text{Bi-linear}_{Q,K,V}(\cdot)$ represents the three bilinear $(\cdot)$ layers for queries Q, keys K, and values V, respectively. The attention score A is then calculated as follows:

$$\text{A} = \frac{\text{Bin-act}(Q) \otimes \text{Bin-act}(K)^T}{\sqrt{D}} \tag{8}$$

where D denotes the feature dimension. Finally, the binarized attention weight B_A^b can be computed as follows:

$$B_A^b = \text{Bin-att}(A). \tag{9}$$

We adhered to the original ViTs architecture to binarize the remaining MSA and MLP layers.

IV. HBiPiDiNet

A. Overview of HBiPiDiNet

Although binarized networks have achieved progress in general object detection, their effectiveness in RSOD remains limited. To enhance RSOD and enable the deployment of models on devices such as drones and satellites, we introduce a binarized network named HBiPiDiNet, specifically designed for this purpose. This network merges two established backbone networks, CNN and ViT, leveraging the advanced binary object detector IDa-Det. A detailed description of this network is provided in Section III. Additionally, we address the underperformance of binarized networks in RSOD tasks by proposing a tailored binarized network, HBiPiDiNet. To address the challenges of weak object response and sparse distribution in remote sensing scene images, we made two innovative attempts: 1) the introduction of BiPDC, which incorporates the classical heuristic algorithm LBP to fully exploit the local contrast priors, thereby enhancing the response of key objects and proposing a parallel version of BiPDC specifically designed for remote sensing images; and 2) the development of HKFCB, which employs a heterogeneous convolutional approach to detect objects in sparsely distributed and variably scaled remote sensing images, thereby further improving performance. The architecture is illustrated in Fig. 2.

B. Binary PDC

The binarization of convolution results in a loss of characteristic responses to key objects, particularly in remote sensing images where the feature response is inherently weak, leading to a diminished representation capability of the network. To enhance the feature response in key local areas of the network, identify regions with prominent edges, and filter out redundant information with less obvious gradients, we introduce the center difference form [22] of PDC [23]

$$y = f(\boldsymbol{x}) = \sum_{i=1}^{9} w_i \cdot x_i, \quad \text{(3*3 vanilla convolution)} \tag{10}$$

$$y = f(\Delta\boldsymbol{x}) = \sum_{i=1, i\neq 5}^{9} w_i \cdot (x_i - q), \quad \text{(3*3 PDC)} \tag{11}$$

where w_i denotes kernel weight. x_i represents the local region pixel and q represents the center pixel.

The central concept of PDC involves calculating the differences between the central pixel and its surrounding pixels. By employing PDC, the model captures a diverse array of contrast patterns within the feature maps. In FPN, feature maps vary in size and semantic depth. The integration of PDC into these networks enriches the feature representation, encompassing not only basic attributes such as color, intensity, and orientation but also local contrast, which is crucial for visual perception. This enhanced representation is particularly beneficial in object detection tasks, where a detailed understanding of the scene is essential. To address efficiency concerns, a binary version of PDC, termed BiPDC, has been introduced

$$y = f(\boldsymbol{x^b}) = \sum_{i=1}^{9} w_i^b \cdot R\text{sign}(x_i - \tau), \quad \text{(vanilla BiConv)} \tag{12}$$

$$y = f((\Delta\boldsymbol{x})^{\boldsymbol{b}}) = \sum_{i=1, i\neq 5}^{9} w_i^b \cdot R\text{sign}(x_i - q), \quad \text{(BiPDC)} \tag{13}$$

where w_i^b represents the binary weights, τ represents the threshold for binarization, and Rsign corresponds to the binarization process of ReactNet. BiPDC differs significantly from BiConv in its processing of features. Although BiConv assigns a binarization result of either $+1$ or -1 based on a threshold from the ReAct-Sign activation function, BiPDC compares the surrounding pixels to the central pixel and assigns $+1$ or -1 accordingly. This method incorporates local contrast priors into the binarization process, which is crucial for RSOD tasks where objects are small and feature responses are subtle. By focusing on local contrast rather than a global threshold, BiPDC more effectively identifies subtle differences in the object area, thereby enhancing the detection of weakly responsive or small-scale features in the image. Although local contrast priors are significant, the recognition and localization capabilities of the neural networks rely fundamentally on the analysis of basic features such as color, shape, and texture. Thus, the PDC at each layer of the FPN can extract rich contrast cues while preserving the extraction of basic features through standard binary convolution operators. In previous BiPDC architectures, the replacement of certain BiConv channels with BiPDC channels prioritized local contrast at the

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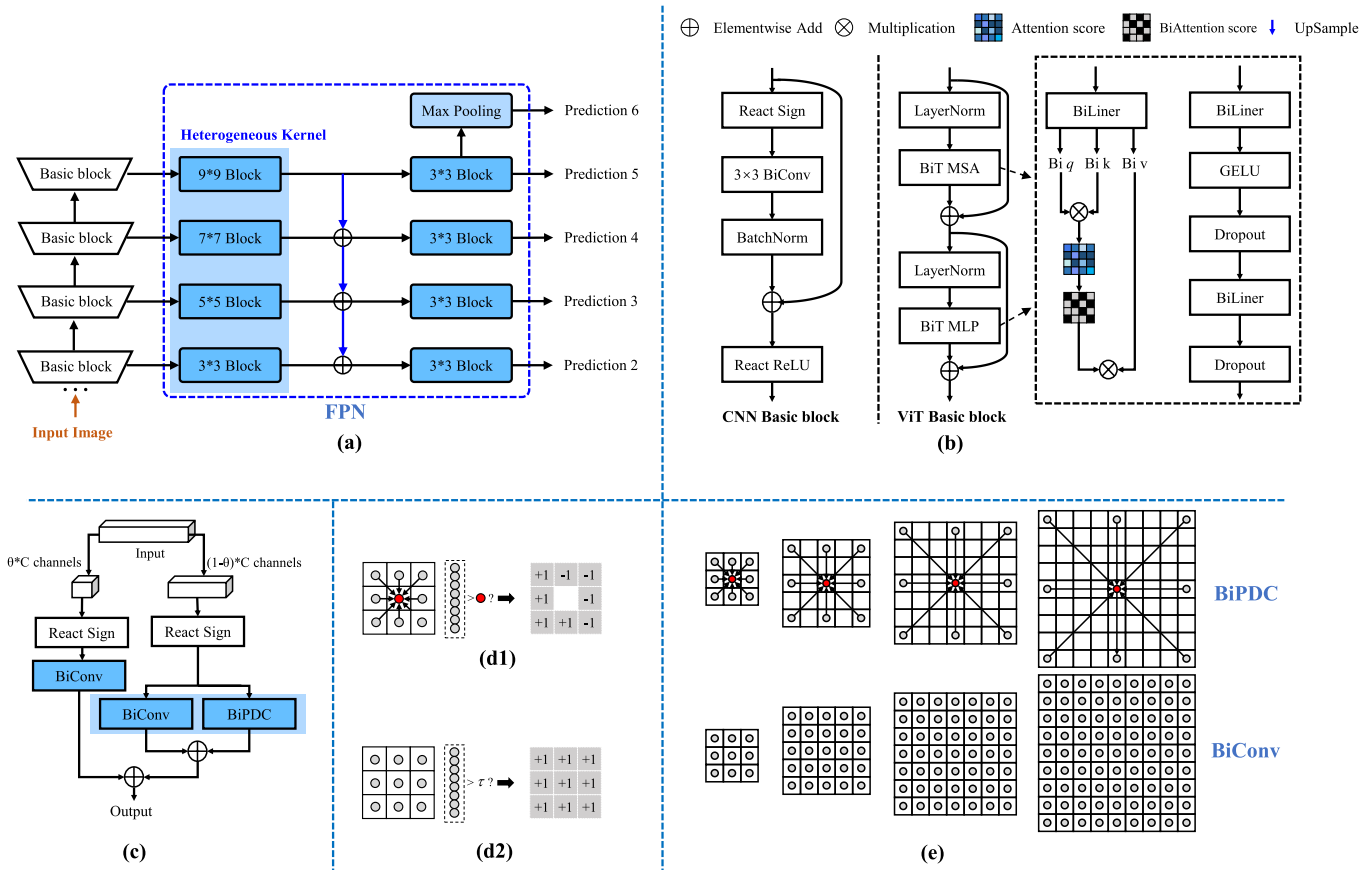


Fig. 2. Illustration of our proposed binary RSOD detection framework HBiPiDiNet. (a) Overall macro architecture of HBiPiDiNet. The structures proposed herein are all embodied within the FPN block. (b) Two types of binarized backbones: CNNs and ViTs. The CNN basic block first undergoes binary activation through React Sign, then extracts features through BiConv, followed by normalization via React ReLU. The entire structure adopts a residual configuration. The ViT basic block primarily consists of binary versions of MSA and MLP, with the specific structures of these two components detailed within dashed boxes. The entire structure also employs a residual configuration. (c) FPN structure, comprising BiPDC and vanilla binary convolution, employs a parallel architecture. (d) Workflow diagram of BiPDC and BiConv. (e) HKFCBs are set to expand the size of the convolution kernels based on the parallel structure of BiPDC.

expense of basic feature representation. To maintain a consistent representation of fundamental features while enhancing local contrast, this study fine-tuned the structure by adding a parallel BiConv branch with an equivalent number of channels to BiPDC. Compared to conventional BiPDC, this approach balances high-level semantic information with local contrast, thereby preserving a more comprehensive feature set during the feature fusion stage

$$y = f((\Delta \boldsymbol{x})^{\boldsymbol{b}} + \boldsymbol{x}^{\boldsymbol{b}}) = \sum_{i=1, i\neq 5}^{9} w_i^b \cdot R\,\mathrm{sign}(x_i - q) + \sum_{i=1}^{9} w_j^b \cdot R\,\mathrm{sign}(x_i - \tau), \quad \text{(Parallel form)}. \tag{14}$$

Although BiPDC can effectively extract the critical local contrast priors, it lacks the inherent capability to capture high-level semantic information. In contrast, BiConv is adept at acquiring high-level semantic details, surpassing BiPDC in this regard. The parallel configuration combines the direct outputs of both BiConv and BiPDC to realize feature fusion. This method adopts a parallel structure of the BiPDC by default and enhances the ability of the network for precise object localization and emphasizes the key local information within object regions. As depicted in Fig. 2(c) and Fig. 3, BiPDC and BiConv operate in parallel in practical scenarios. This strategy involves employing a parallel BiPDC operator at each layer of the neural network architecture to extract rich, multilevel, and multiscale contrastive visual clues while preserving the high-frequency information. Simultaneously, standard binary convolution is crucial for maintaining basic visual features such as color, contours, and texture. FPN blocks are partitioned along the channel into BiConv and BiPDC components, with features from the same layer subsequently merged. The merging is governed by the following formula:

$$\begin{aligned} y &= f((\Delta \boldsymbol{x})^{\boldsymbol{b}} + \boldsymbol{x}^{\boldsymbol{b}}, \theta) \\ &= (1-\theta) * \sum_{i=1}^{9} w_j^b \cdot R\,\mathrm{sign}(x_i - \tau) + \theta \\ &\quad * \left(\sum_{i=1, i\neq 5}^{9} w_i^b \cdot R\,\mathrm{sign}(x_i - q) + \sum_{i=1}^{9} w_j^b \cdot R\,\mathrm{sign}(x_i - \tau) \right), \quad \text{(FPN block)} \end{aligned} \tag{15}$$

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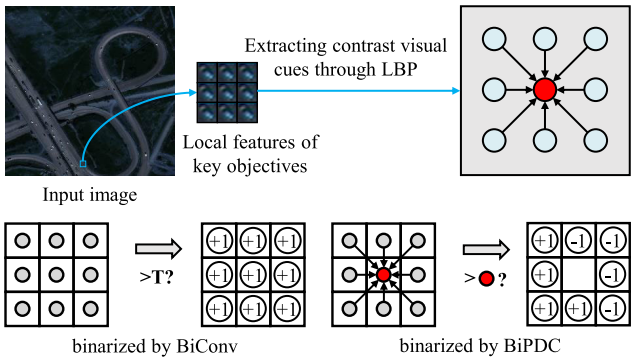


Fig. 3. In conventional BiConv, pixels in small object areas of remote sensing images are easily binarized to the same value, as the threshold is optimized for the entire input rather than specific local regions. This finding represents the loss of valuable local contrast prior information. In contrast, BiPDC employs neighboring pixels as the threshold, utilizing the principles of LBP. By altering the form of binary convolution operations, BiPDC flexibly enhances the representational capability. This approach preserves crucial local contrast information and ensures that pixels within a region are less likely to be binarized into identical values, thereby significantly enhancing the detection of subtle features in remote sensing images.

where the hyperparameter θ balances the contribution between contrastive and basic visual clues, with larger values of θ implying a greater emphasis on contrastive visual clues.

C. Heterogeneous Kernel Fusion Convolution Blocks (HKFCB)

Multiscale feature extraction is critical in remote sensing image processing [68], [69]. The prevalence and nonuniform distribution of small objects in remote sensing scenes highlight the importance of multiscale feature extraction capabilities in RSOD models. Extracting scale information from remote sensing scenes presents a significant challenge for binary networks. To effectively integrate the varying scale information of multiple objects in such scenes, we address the issue of feature fusion using homogeneous convolution within the FPN of binary object detectors. In a pyramidal structure, high-resolution features from the shallow stages of the detector are utilized to capture fine-grained semantics for detecting small objects, whereas low-resolution features from deeper stages are employed to capture higher-level semantics for detecting larger objects. The conventional pyramid architecture, which employs uniform small-kernel convolutions across all stages, limits the effective receptive field in deeper stages, and therefore, impairs the detector to perceive scale in remote sensing scenes. Large-kernel convolutions with expansive receptive fields can encode broader areas [70], [71], [72]. Despite the predominance of model binarization, the heterogeneous kernel convolution structure only slightly increases computational complexity compared to the homogeneous convolution structure. Thus, we propose a convolutional block that combines binary convolution with BiPDC, modifying the size of convolutional layers to address the demands of multiscale feature information extraction. Given the sparse distribution characteristics of objects in remote sensing images, the heterogeneous fusion convolutional block extends the initially designed 3×3 binary convolution and 3×3 BiPDC to $k \times k$ (where $k = 3, 5, 7, 9$). The formula used is as follows:

$$y = f(\boldsymbol{x}^{\boldsymbol{b}}) = \sum_{i=1}^{k*k} w_i^b \cdot R\text{sign}(x_i - \tau), \quad \text{(heterogeneous BiConv)} \quad (16)$$

$$y = f((\Delta \boldsymbol{x})^{\boldsymbol{b}}) = \sum_{i=1, i \neq \frac{k*k+1}{2}}^{k*k} w_i^b \cdot R\text{sign}(x_i - q), \quad \text{(heterogeneous BiPDC).} \quad (17)$$

A convolution with a 3×3 kernel operates in the first stage of the FPN. Thereafter, the kernel size is gradually increased in the intermediate stages and aligned with the incremental feature resolution. A large 9×9 kernel is employed in the final stage of the FPN. This design provides heterogeneous binary convolutions and BiPDC with multiple kernel sizes in the deeper stages. HKFCB operates on the FPN to adjust the size of its convolutional kernels at each layer using the following formula:

$$\begin{aligned} y &= f((\Delta \boldsymbol{x})^{\boldsymbol{b}} + \boldsymbol{x}^{\boldsymbol{b}}, \theta) \\ &= (1-\theta) * \sum_{i=1}^{k*k} w_j^b \cdot R\,\text{sign}(x_i - \tau) + \theta \\ &\quad * \left(\sum_{i=1, i \neq k}^{k*k} w_i^b \cdot R\,\text{sign}(x_i - q) \right. \\ &\quad \left. + \sum_{i=1}^{k*k} w_j^b \cdot R\,\text{sign}(x_i - \tau) \right), \quad \text{(HKFCB).} \end{aligned} \quad (18)$$

The $k \times k$ BiPDC enhances the single-scale LBP operator by extending it to multiple scales. This scaling facilitates the extraction of edge features from objects that are widely distributed in remote sensing images. By integrating the large receptive field of binary convolutions, the BiPDC enriches the high-level semantic information related to object distribution in these images, thereby improving the multiscale representation capability of the network for nonuniform distributions. High-level feature maps are relatively abstract and typically encompass broad contextual information. In contrast, low-level feature maps contain more specific details, focusing primarily on texture and finer attributes. In RSOD, high-level feature maps are crucial for processing global and abstract features. Remote sensing images often cover vast geographic areas, including cities, farmlands, and forests, requiring the capture of relationships between these regions and their overall layout. For large-scale targets and complex landscapes, global context is essential for precise detection. Employing large convolutional kernels in the upper layers of feature pyramids helps the model capture macro features and aids in learning the complex contextual relationships between the target and its background by covering larger areas. On the other hand, the need for low-level feature maps is to process small local features. Small-sized targets in remote sensing images are effectively recognized by capturing detailed local features, where small convolutions focus on extracting these details without losing crucial spatial information. In RSOD, using large convolutions in the higher layers and small convolutions in the lower layers

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provides the high-level feature maps with the capability to handle global context and large-scale information processing, while supplementing the low-level maps with rich detail and precise spatial information. This design enables the model to maintain efficiency and accuracy when processing information from micro to macro scales, making it particularly suitable for handling the high complexity and multiscale features of remote sensing scenes [73].

V. Experiments

A. Datasets

The experiments were conducted using three datasets. The primary experiments utilized the AI-TOD dataset [74], which is challenging due to its small average absolute object size of 12.8 pixels and consists of 14 018 training images and an equal number of testing images. Furthermore, the proposed method was evaluated using the VisDrone2019 [75] and DIOR datasets [76], which are commonly employed in remote sensing research.

B. Experiment Settings

All experiments were performed on an NVIDIA RTX 3090 GPU. Model training was conducted using the PyTorch framework within the MMdetection environment [77]. For the CNN-based detector, ResNet18 [78] served as the backbone, while for the transformer-based detector, the backbones were composed of PvT [79], PoolFormer [80], Swin [81], and ConvNext [82], all of which were pre-trained on ImageNet. The models were trained using the stochastic gradient descent (SGD) optimizer for 12 epochs, with a momentum of 0.9, weight decay of 0.0001, batch size of 4, and an initial learning rate of 0.004. The learning rate was adjusted to 0.0001 when using the AdamW optimizer and was reduced by a factor of 0.1 at epochs 8 and 11. The number of RPN proposals was set to 3000. During inference, the confidence score threshold was set at 0.05 to eliminate the background bounding boxes, and the NMS IoU threshold was set at 0.5, considering the first 3000 bounding boxes. All other parameters adhered to the default settings in MMdetection.

C. Evaluation Metrics

To assess the model performance on remote sensing datasets, we described the performance indicators used. Average precision (AP) and mean AP (mAP) serve as evaluation metrics and are commonly employed in object detection studies involving natural and remote sensing images. The AP is calculated by measuring the area under the precision–recall curve, which spans recall rates from 0% to 100%. Precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{20}$$

The predicted bounding boxes with an intersection over union (IoU) score above a predefined threshold are classified as either true positives (TPs) or FPs. FNs refer to undetected ground-truth bounding boxes. The AP values fluctuate with different IoU thresholds, generally decreasing as the IoU threshold increases. A higher AP at a specific IoU threshold indicates superior performance

$$\text{AP} = \frac{\sum_{k=1}^{n} (P(k) \times r(k))}{|R(q)|}$$

$$\text{mAP} = \frac{1}{Q} \sum_{q=1}^{Q} \text{AP}(q) \tag{21}$$

where Q represents the number of categories, $|R(Q)|$ denotes the number of images associated with category Q, k is the rank in the retrieved image sequence, n denotes the total number of retrieved images, $P(k)$ indicates the precision at cutoff point k in the list, and $R(q)$ represents an indicator function that assigns a value of 1 if the image at rank q is relevant, and 0 otherwise. mAP_{50} refers to the scenario where the IoU threshold for calculating mAP is 0.5. IoU quantifies the overlap between predicted and true bounding boxes. A prediction is deemed correct when IoU ≥ 0.5. Thus, mAP_{50} adopts a more lenient matching criterion, primarily assessing whether the object has been approximately detected. Conversely, mAP_{75} employs an IoU threshold of 0.75, representing a stricter criterion that demands greater alignment between predicted and true bounding boxes. mAP_{75} rigorously evaluates the accuracy of the model in object localization. The AI-TOD dataset includes aerial objects of varying sizes. Objects are categorized by size as follows: very small objects range from $[2, 8]$ pixels, small objects from $[8, 16]$ pixels, smaller objects from $[16, 32]$ pixels, and medium-sized objects from $[32, 64]$ pixels. There are no large objects in this dataset. The distribution of these categories is 13.3%, 72.3%, 12.3%, and 2.1%, respectively. The experimental section evaluates these object sizes using the AP metrics. To measure the complexity of the BNN [55], [83], we employ total operations (OPs) as follows:

$$\text{OPs} = \frac{\text{BOPs}}{64} + \text{FLOPs}. \tag{22}$$

BOPs quantify the computational load of binary operations, whereas FLOPs account for the load of real-valued operations. Since our proposed method introduces no new binary operations, the calculation method remains consistent with that used in Reactnet [55] and BiT [58]. The marginal increase in OPs primarily originates from additional FLOPs.

D. Main Results on AI-TOD

Table I compares our method with SOTA binary detectors on the AI-TOD benchmark, including BiDet and IDa-Det. We employed Faster RCNN and Retinanet as detection frameworks. The backbones used were ResNet18 [78], Poolformer-s12 [80], PVT-t [79], Swin-t [81], and ConvNext-t [82]. Owing to the low accuracy of single-stage detectors, their real-valued counterparts yielded unsatisfactory results. We utilized only ResNet18 to validate the effectiveness of our method on a single-stage detector. To verify the compatibility and performance of the proposed method, we incorporated RFLA [67], which is a label allocation method specifically devised for RSOD. The use of RFLA, indicated by "*", involved its

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TABLE I

MAIN RESULTS ON AI-TOD. NOTE THAT MODELS ARE TRAINED ON THE TRAINVAL SET AND VALIDATED ON THE TEST SET. "*" INDICATES RFLA. NOTE THAT THE ORIGINAL IMPLEMENTATION OF BIDET DOES NOT CONTAIN AN FPN LAYER, RESULTING IN POOR PERFORMANCE ON RSOD. WE FOLLOWED THE BIDET QUANTIZATION METHOD AND ADDED BINARY FPN FOR ITS PERFORMANCE IMPROVEMENT. FOR IDA-DET, WE SIMPLIFIED THE TEDIOUS PROCESS OF KNOWLEDGE DISTILLATION AND USED THE SAME IMPLEMENTATION FOR EXPERIMENTS ON OTHER DATASETS

Framework	Backbone	Quantization Method	W/A	mAP	mAP_{50}	mAP_{75}	mAP_{vt}	mAP_t	mAP_s	mAP_m	OPs($\times 10^9$)
Faster RCNN	ResNet-18	Real-Value	32/32	10.9	25.9	7.4	0	8.0	21.9	30.9	101.42
		Real-Value*		18.3	45.7	10.9	7.5	19.0	22.5	28.7	101.42
		IDa-Det	1/1	7.7	19.3	4.7	0	6.6	14.9	24.0	15.77
		IDa-Det*		12.2	32.3	6.4	2.5	12.1	16.4	21.8	15.77
		BiDet		2.6	7.0	1.6	0	2.2	5.4	7.4	17.03
		BiDet*		4.6	12.9	2.0	0.6	4.8	5.3	6.5	17.03
		HBiPiDiNet		8.4(0.7 ↑)	20.9(1.6 ↑)	5.3(0.6 ↑)	0	7.0(0.4 ↑)	16.4(0)	26.5(2.5 ↑)	16.34(0.57 ↑)
		HBiPiDiNet*		14.0(1.8 ↑)	36.5(4.2 ↑)	8.0(1.6 ↑)	4.2(1.7 ↑)	14.6(2.5 ↑)	17.4(1.0 ↑)	24.6(2.8 ↑)	16.34(0.57 ↑)
	Poolformer-s12	Real-Value	32/32	12.9	30.3	9.0	0	9.1	25.7	34.2	101.41
		Real-Value*		23.5	56.1	15.2	10.6	24.2	28.2	33.3	101.41
		IDa-Det	1/1	7.0	17.0	4.5	0	5.0	13.7	22.2	18.32
		IDa-Det*		11.2	29.6	6.1	1.8	11.0	14.5	20.7	18.32
		BiDet		6.3	15.5	3.9	0	5.1	12.5	19.0	19.39
		BiDet*		10.8	28.1	6.0	2.3	11.1	13.1	16.9	19.39
		HBiPiDiNet		7.7(0.7 ↑)	18.6(1.6 ↑)	5.1(0.6 ↑)	0	5.5(0.5 ↑)	15.8(2.1 ↑)	22.5(0.3 ↑)	18.92(0.6 ↑)
		HBiPiDiNet*		12.0(0.8 ↑)	30.8(0.8 ↑)	7.0(0.9 ↑)	2.2(0.4 ↑)	11.8(0.8 ↑)	15.8(1.3 ↑)	21.0(0.3 ↑)	18.92(0.6 ↑)
	PVT-t	Real-Value	32/32	10.8	25.4	7.3	0	7.7	22.5	31.1	114.27
		Real-Value*		19.2	47.8	11.5	9.1	20.0	22.0	29.7	114.27
		IDa-Det+BiT	1/1	4.8	11.6	3.0	0	3.6	9.1	15.4	17.02
		IDa-Det+BiT*		8.5	21.8	4.6	1.2	8.4	10.3	15.0	17.02
		BiDet+BiT		1.2	3.4	0.7	0	1.3	2.1	2.9	18.08
		BiDet+BiT*		3.2	9.4	1.3	0.4	3.9	3.0	2.6	18.08
		HBiPiDiNet		5.3(0.5 ↑)	13.5(1.9 ↑)	3.5(0.5 ↑)	0	3.9(0.3 ↑)	10.2(1.1 ↑)	15.9(0.5 ↑)	17.62(0.6 ↑)
		HBiPiDiNet*		9.1(0.6 ↑)	23.7(1.9 ↑)	5.0(0.4 ↑)	2.3(1.1 ↑)	9.0(0.6 ↑)	11.7(1.4 ↑)	17.1(2.1 ↑)	17.62(0.6 ↑)
	Swin-t	Real-Value	32/32	13.8	32.8	9.6	0	10.8	26.1	35.9	137.78
		Real-Value*		23.5	56.5	15.0	9.9	23.9	28.7	35.1	137.78
		IDa-Det+BiT	1/1	6.8	15.9	4.4	0	4.7	12.9	21.7	20.21
		IDa-Det+BiT*		10.1	26.1	5.5	1.9	9.9	12.6	19.2	20.21
		BiDet+BiT		6.1	14.6	4.1	0	4.4	11.6	19.0	21.47
		BiDet+BiT*		10.5	26.5	6.1	2.0	10.1	12.9	20.0	21.47
		HBiPiDiNet		7.1(0.3 ↑)	16.7(0.8 ↑)	4.8(0.4 ↑)	0	5.0(0.3 ↑)	14.5(1.6 ↑)	22.2(0.5 ↑)	20.99(0.78 ↑)
		HBiPiDiNet*		12.4(2.3 ↑)	31.1(5.0 ↑)	7.2(1.7 ↑)	2.8(0.9 ↑)	12.4(2.5 ↑)	15.7(3.1 ↑)	20.5(1.3 ↑)	20.99(0.78 ↑)
	ConvNext-t	Real-Value	32/32	13.3	31.6	9.3	0	10.6	25.7	36.1	135.65
		Real-Value*		21.8	55.0	12.7	10.1	21.9	27.7	35.7	135.65
		IDa-Det+BiT	1/1	5.5	13.4	3.6	0	4.1	11.1	18.8	19.97
		IDa-Det+BiT*		9.3	23.7	5.2	1.5	9.1	11.5	15.5	19.97
		BiDet+BiT		5.1	12.1	3.3	0	3.8	10.2	17.6	20.59
		BiDet+BiT*		8.3	21.5	4.6	1.7	8.2	10.4	16.3	20.59
		HBiPiDiNet		6.1(0.6 ↑)	14.6(1.2 ↑)	4.1(0.5 ↑)	0	4.5(0.4 ↑)	12.7(1.6 ↑)	19.8(1.0 ↑)	20.20(0.23 ↑)
		HBiPiDiNet*		10.2(0.9 ↑)	26.4(2.7 ↑)	5.6(0.4 ↑)	3.0(1.5 ↑)	10.2(1.1 ↑)	12.4(0.9 ↑)	18.2(2.7 ↑)	20.20(0.23 ↑)
Retinanet	ResNet-18	Real-Value	32/32	5.1	15.0	2.3	1.5	4.7	9.5	15.3	98.12
		Real-Value*		7.6	21.0	3.8	2.5	8.3	11.6	14.0	98.12
		IDa-Det	1/1	3.1	10.0	1.2	1.0	2.8	5.7	11.5	66.98
		IDa-Det*		4.0	13.1	1.2	1.2	4.3	6.5	9.0	66.98
		BiDet		1.2	4.5	0.4	0.6	1.3	1.5	2.3	72.0
		BiDet*		2.1	7.1	0.5	0.6	2.4	2.9	1.8	72.0
		HBiPiDiNet		3.7(0.6 ↑)	11.3(1.3 ↑)	1.4(0.2 ↑)	1.1(0.1 ↑)	3.3(0.5 ↑)	5.8(0.1 ↑)	11.9(0.4 ↑)	67.58(0.6 ↑)
		HBiPiDiNet*		4.5(0.5 ↑)	14.2(1.1 ↑)	1.6(0.4 ↑)	1.4(0.2 ↑)	4.7(0.4 ↑)	7.8(1.3 ↑)	10.5(1.5 ↑)	67.58(0.6 ↑)

application as an anchor box assignment and training strategy. Based on the experimental results, the following conclusions were inferred. The binarization process results in information loss that degrades the ability of the model to accurately represent the critical visual cues such as color, texture, and shape of remote sensing objects. Utilizing the binarization approach of BiRealNet, BiDet experienced a significant decline in performance. Conversely, IDa-Det, employing the binarization technique from ReactNet and modifying the activation distribution through channel threshold adjustments, mitigated this performance degradation. To address the issues of low object feature responsiveness and sparse object distribution in RSOD, we integrated BiPDC and HKFCB into a binary FPN based on IDa-Det. BiPDC introduces a local contrast prior, enhancing key object responsiveness, whereas HKFCB improves scale-aware capabilities, thereby improving overall performance.

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TABLE II
MORE BINARIZATION DETECTOR RESULTS (ALL RESULTS WERE OBTAINED USING RFLA)

Quantization Method	W/A	mAP	mAP_{50}	mAP_{75}	mAP_{vt}	mAP_t	mAP_s	mAP_m
Real-value	32/32	18.3	45.7	10.9	7.5	19.0	22.5	28.7
LSQ	4/4	13.4	34.4	8.0	3.8	13.9	16.7	21.8
IDa-Det	1/1	12.2	32.3	6.4	2.5	12.1	16.4	21.8
Adabin	1/1	6.2	15.9	3.7	0.8	5.5	8.4	13.0
ReBNN	1/1	12.6	32.9	6.8	2.6	12.7	16.7	22.0
ReCU	1/1	7.8	20.4	4.5	1.1	7.6	10.4	15.3
IRNet	1/1	2.0	6.5	0.7	0.4	2.5	2.5	0.7
HBiPiDiNet	1/1	14.0	36.5	8.0	4.2	14.6	17.4	24.6

E. Extension on More Binarized Object Detectors on AI-TOD

Binary detection models reduce the number of network parameters and the level of computational demands, which also results in information loss and notably affects the key local responses in RSOD scenarios. To assess the reliability of our methods, we compared various quantization techniques to ascertain their impact on the effectiveness of our approach, thereby demonstrating its advanced nature and effectiveness. In Table II, our method is compared with prominent binarization approaches including LSQ [84], Adabin [85], ReBNN [86], ReCU [87], and IRNet [88]. In particular, LSQ introduces a novel method for estimating and adjusting quantizer step sizes for each weight and activation layer based on the task loss gradient, facilitating concurrent learning with other network parameters. This method employs varying precision levels tailored to system requirements, with a 4-bit version used for our experiments. Adabin [85] proposes an adaptive quantization function that searches for the optimal binary set for weights and intermediate features at each layer, optimizing the binarization process for both weights and activations to minimize the feature representation gap between binary and floating-point formats. ReBNN [86] addresses the weight fluctuations during the training of BNNs, which can impair the representational capabilities. It introduces parameterized scaling factors and a weighted reconstruction loss to establish an adaptive training objective, effectively controlling weight oscillations with a balance parameter linked to the reconstruction loss. ReCU examines the impact of "dead weights," which are weights that seldom update during BNN training. It introduces a rectified clamping unit (ReCU) to reactivate these "dead weights" for updates, thus reducing quantization errors. IRNet enhances network accuracy through information restriction and recovery, selecting information-rich shallow feature maps and fusing them with final feature maps to restore extracted feature information, thereby matching the input information amount. The experimental results reveal that our proposed HBiPiDiNet achieves higher accuracy compared to the selected mainstream binarization methods and even the 4-bit LSQ. These results reveal that the mainstream 4-bit method, LSQ, and several mainstream 1-bit methods significantly reduce accuracy compared to real-valued networks, challenging their adaptability to RSOD task characteristics. Moreover, HBiPiDiNet exceeds all compared quantization methods, validating the effectiveness and adaptability of the HBiPiDiNet binarization object detection method in RSOD.

F. Results on More Datasets

To assess the generalizability of our method across remote sensing datasets, we conducted additional experiments on VisDrone2019 and DIOR using ResNet18. Training was performed on the trainval set of each dataset, with performance subsequently evaluated on the test set. The results are detailed in Table III. On the VisDrone2019 dataset, the mAP of HBiPiDiNet improved by 0.9% and 1.0% compared to that of IDa-Det. On the DIOR dataset, improvements of 4.0% and 6.2% were observed. HBiPiDiNet consistently demonstrated significant enhancements across all metrics on both datasets.

G. Ablation Study

In the ablation studies, we employed a Faster RCNN with ResNet18 as the backbone and RFLA as the labeling strategy by default.

1) Effectiveness of Individual Components: Table IV presents ablation studies analyzing the components BiPDC and HKFCB separately. The introduction of the original BiPDC form increased the mAP by 1.0%. Implementing parallel BiPDC further raised the mAP by 1.2%. The addition of HKFCB to the parallel BiPDC configuration resulted in a mAP increase of 1.8% compared to IDa-Det. Based on these results, we can conclude that: 1) BiPDC is the primary driver of performance enhancement, serving as the focal point of this study. As a novel binary convolution technique, it incorporates classical heuristic algorithms into the convolution process, leveraging local contrast prior to enhance the feature response of key objects and address the issue of weak feature response in RSOD scenarios. Consequently, it significantly boosts the performance of binary networks in RSOD tasks. 2) HKFCB expands BiPDC in the FPN into a heterogeneous structure, which introduces scale awareness to the model, allowing it to accommodate varying object scales in remote sensing contexts and further improve performance. 3) The integration of these methods substantially enhances the performance of binary networks in RSOD tasks while only marginally increasing the computational load, thus achieving a balance between accuracy and efficiency.

2) Impact of θ of BiPDC: The hyperparameter θ in the BiPDC component was adjusted to balance the contributions between contrastive visual cues and basic visual semantic cues. Fig. 4 illustrates how the detection mAP varies with different θ values across three datasets. As observed, the optimal performance is achieved at $\theta = 0.2$. Deviations from this value, either higher or lower, result in decreased mAP, indicating that an appropriate θ value effectively balances contrasting visual cues and basic visual semantic cues for optimal performance. Furthermore, the trend in mAP changes with θ values is consistent across the AI-TOD, VisDrone2019, and DIOR datasets, suggesting that the optimal θ value is independent of the dataset. Additionally, when θ ranged between 0.1 and 0.3, the variation in detected mAP was minimal, indicating some tolerance of the model to changes in the value of θ.

3) Impact of Composition of BiPDC and BiConv: BiPDC was combined with BiConv for performance optimization.

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TABLE III
RESULTS ON VISDRONE2019 AND DIOR. THE SYMBOL "*" MEANS WITH RFLA

Quantization Method	VisDrone2019				DIOR			
	mAP	mAP_{50}	mAP_{vt}	mAP_t	mAP	mAP_{50}	mAP_{vt}	mAP_t
Real-Value	15.6	29.4	0.1	11.7	38.9	66.2	0.0	14.3
IDa-Det	13.5	25.6	0.7	9.7	26.7	50.9	0.0	9.7
HBiPiDiNet	14.4(0.9 ↑)	27.0(1.4 ↑)	0.1(0.6 ↓)	11.3(1.6 ↑)	30.7(4.0 ↑)	56.7(5.8 ↑)	0.0	10.4(0.7 ↑)
Real-Value*	16.4	32.6	2.2	12.6	40.0	68.1	0.8	15.1
IDa-Det*	13.7	27.7	1.5	10.4	27.3	51.9	0.3	9.7
HBiPiDiNet*	14.7(1.0 ↑)	29.6(1.9 ↑)	2.6(1.1 ↑)	12.3(1.9 ↑)	33.5(6.2 ↑)	60.8(8.9 ↑)	1.0(0.7 ↑)	13.3(3.6 ↑)

TABLE IV
EFFECTIVENESS OF INDIVIDUAL COMPONENTS

	mAP	mAP_{50}	mAP_{75}	mAP_{vt}	mAP_t	mAP_s	mAP_m	$OPs(\times 10^9)$
IDa-Det	12.2	32.3	6.4	2.5	12.1	16.4	21.8	15.77
+ BiPDC (original form)	13.2(1.0 ↑)	34.2(1.9 ↑)	7.4(1.0 ↑)	2.8(0.3 ↑)	13.7(1.6 ↑)	17.1(0.7 ↑)	22.3(0.5 ↑)	15.97(0.20 ↑)
+ BiPDC (parallel form)	13.4(1.2 ↑)	34.7(2.4 ↑)	7.5(1.1 ↑)	3.5(1.0 ↑)	14.0(1.9 ↑)	17.1(0.7 ↑)	23.1(1.3 ↑)	16.01(0.24 ↑)
+ BiPDC + HKFCB	14.0(1.8 ↑)	36.5(4.2 ↑)	8.0(1.6 ↑)	4.2(1.6 ↑)	14.6(2.5 ↑)	17.4(1.0 ↑)	24.6(2.8 ↑)	16.34(0.57 ↑)

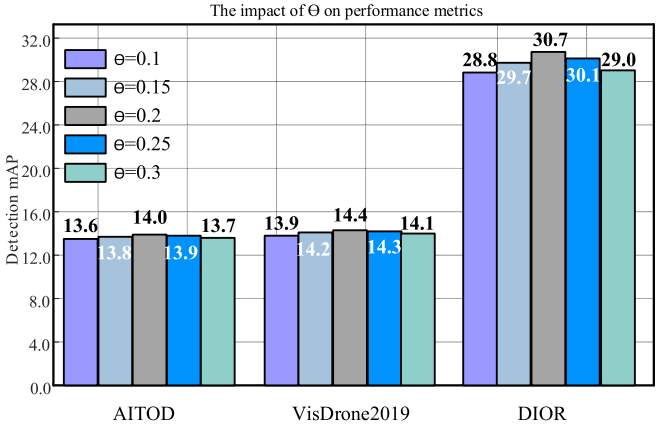


Fig. 4. Theta regulates the proportion of channels occupied by BiPDC and determines the trade-off between contrast prior and high-level semantic information, thereby affecting the accuracy of the model. The data exhibits the performance under different θ values when other conditions remain unchanged. As the θ values of BiPDC, we selected 0.1, 0.15, 0.2, 0.25, and 0.3. In the experiments, both excessively high and low values resulted in performance degradation.

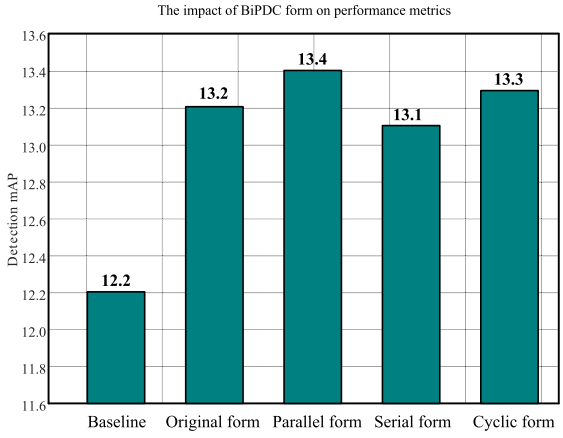


Fig. 5. Results reveal the impact of the different BiPDC composition forms on the detection of mAP in AI-TOD. These five detection mAPs correspond to the five BiPDC combination forms in Fig. 6, indicating that different BiPDC combination forms significantly impact the detection mAP.

The experiments described herein aimed to identify the most effective combination of these methods. As depicted in Fig. 5, we evaluated the impact of original, parallel, serial, and cyclic configurations on HBiPiDiNet. The various configurations are illustrated in Fig. 6. The analysis reveals that: 1) all four configurations incorporating BiPDC significantly outperformed the baseline, confirming that the introduction of BiPDC substantially enhances RSOD; 2) the different BiPDC configurations influenced the detection mAP, with the parallel configuration yielding the best results; and 3) the detection mAP for the serial configuration was lower than that of the original, indicating that overly complex combinations can degrade the integration of contrast and basic visual semantic cues, thereby underscoring the significance of selecting the appropriate configuration.

4) Impact of the Kernel Size of HKFCB: The use of heterogeneous kernel sizes facilitates the extraction of scale information, and the pattern of kernel size variation impacts experimental performance. The kernel sizes for the four stages of the FPN block are denoted as x_1, x_2, x_3, and x_4. Table V presents the variations in detection mAP resulting from different combinations of kernel sizes (x_1, x_2, x_3, x_4) in the FPN. To isolate the influence of other variables, we maintained all other model parameters at their optimal settings. The findings indicate that: 1) employing uniformly sized convolution kernels with increasing sizes results in a modest increase in detection mAP, though the improvement is marginal and incurs additional computational costs; 2) although the use of heterogeneous convolution kernels also increases computational demands, an appropriately configured heterogeneous combination significantly enhances performance; and 3) the optimal kernel size configuration was $x_1 = 3$, $x_2 = 5$, $x_3 = 7$, and $x_4 = 9$.

H. Computational Complexity

To comparatively assess the computational complexity of HBiPiDiNet in comparison to other models, such as the OPs and parameters, we refer to Fig. 7. The experimental

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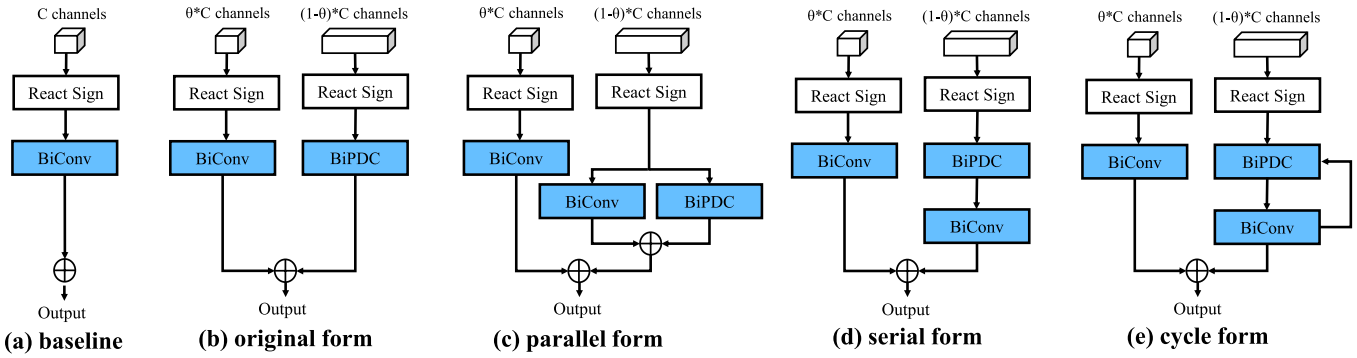


Fig. 6. (a) Baseline without using BiPDC, and (b)–(e) four types of BiPDC combinations in the experiments.

TABLE V

DETECTION PERFORMANCE VARIES WITH THE CHANGE IN HETEROGENEOUS KERNEL SIZES, WITH DATA DISPLAYING KERNEL SIZES ALL SET TO 3, HOMOGENEOUS LARGE KERNEL CONVOLUTIONS, AND VARIOUS HETEROGENEOUS CONVOLUTION FORMS TO EXPLORE THE OPTIMAL KERNEL SIZE SETTING

$[x_1, x_2, x_3, x_4]$	mAP	mAP_{50}	mAP_{vt}	mAP_t	OPs($\times 10^9$)
$[3, 3, 3, 3]$	13.4	34.7	3.5	14.0	16.01
$[5, 5, 5, 5]$	13.7	35.4	3.8	14.2	16.36
$[7, 7, 7, 7]$	13.6	35.2	3.7	14.2	16.61
$[9, 9, 9, 9]$	13.6	35.3	3.6	14.0	17.14
$[9, 7, 5, 3]$	13.5	34.9	3.6	14.2	16.89
$[3, 7, 5, 9]$	13.8	35.8	4.0	14.4	16.44
$[5, 7, 9, 3]$	13.7	35.4	3.8	14.3	16.51
$[3, 5, 7, 9]$	14.0	36.5	4.2	14.6	16.34

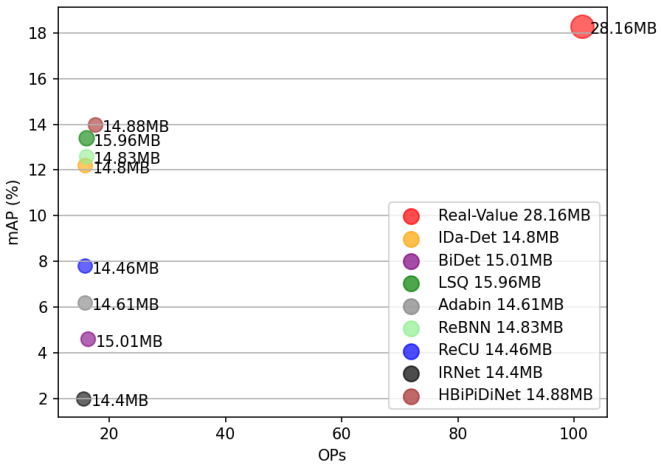


Fig. 7. When using Resnet18 as backbone, faster RCNN as detector, and RFLA as anchor box assignment strategy, the mAP, GFLOPs, and parameter amount corresponding to each binarization method.

results show that HBiPiDiNet slightly increases the number of parameters and computational costs, which reduces detection efficiency compared to other binarized networks. However, these increases are minimal when compared to real-valued networks. Combining the previous experimental results, HBiPiDiNet achieves commendable detection performance in remote sensing scenarios, effectively balancing accuracy and efficiency.

I. Theoretical Power Consumption in Edge Devices

BNNs are ideally suited for deployment on edge devices to fully leverage their benefits. However, the availability of chips capable of operating BNNs is currently limited. Due to experimental constraints, we theoretically evaluate the power consumption of our proposed method, based on Faster R-CNN and ResNet18, on a chip and compare it with real values and IDa-Det binary counterparts. We used Senputing [89] image sensor chip for the simulation. This architecture capitalizes on the photocurrent from the photodiode in a traditional 3T-APS active pixel unit through an analog circuit design. It computes the first layer of convolutional feature maps of the BNN directly within the image sensor and outputs them to the subsequent network for recognition. This requires only a minimal amount of binary feature map data to be externally output, significantly reducing the high energy consumption typically associated with high-precision image output from the ADC in current intelligent vision systems. Utilizing the TSMC 65-nm CMOS process for the intelligent sensing image sensor chip tape-out, the efficiency is measured at 17.3 TOp/W · s. Using the following formula, we calculate the energy consumption per operation (OPs):

$$P = \frac{1}{17.3\ \text{TOP/W} \cdot \text{s}} = 5.78 * 10^{-14}\ \text{J/op}. \tag{23}$$

Based on the OPs from previous experiments, the real value, IDa-Det, and HBiPiDiNet yielded OPs of 101.42×10^9, 15.77×10^9, and 16.34×10^9, respectively. The corresponding power consumptions were calculated as 5.86, 0.91, and 0.94 mJ. The simulation results from theoretical calculations suggest that binarization significantly reduces the power consumption of networks on edge devices. The HBiPiDiNet demonstrated an improvement in accuracy while maintaining similar power consumption to that of IDa-Det.

J. Extension on BiPDC to Backbone in AI-TOD

As indicated in Table VI, the proposed method, applied to the FPN part of the detector, exhibited good performance. To assess its scalability, experiments were conducted to determine if the parallel form of BiPDC, when replacing the binary convolutions in the backbone, could yield favorable outcomes. These experiments employed ResNet18 as the backbone and RFLA as the assignment strategy. To ensure fairness,

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TABLE VI
REGARDING THE EXPERIMENTAL COMPARISON OF EXTENDING THE METHOD OF THIS ARTICLE TO THE BACKBONE

Backbone	FPN	mAP	mAP_{50}	mAP_{75}	mAP_{vt}	mAP_t	mAP_s	mAP_m
Real-value	Real-value	18.3	45.7	10.9	7.5	19.0	22.5	28.7
IDa-Det	IDa-Det	12.2	32.3	6.4	2.5	12.1	16.4	21.8
IDa-Det	HBiPiDiNet	14.0	36.5	8.0	4.2	14.6	17.4	24.6
HBiPiDiNet	IDa-Det	9.5	25.0	5.2	1.8	9.4	12.7	17.0
HBiPiDiNet	HBiPiDiNet	10.9	28.1	6.2	2.4	11.1	13.9	18.0

two sets of extended experiments were conducted. In these experiments, the FPN was configured either as IDa-Det or HBiPiDiNet, with the backbone convolutions replaced by BiPDC. Specifically, all 3×3 binary convolutions in the basic block of the backbone were replaced with a parallel form of BiPDC. However, this modification resulted in decreased detection accuracy, suggesting that the use of BiPDC in the backbone is not advisable. The potential reasons are twofold. as listed below.

1) Although the parallel form of BiPDC in the FPN can enhance the feature response in key local areas, thereby optimizing the output of subsequent stages, its use in the backbone may disrupt the feature response distribution during the feature extraction stage. A highly concentrated response does not facilitate optimal feature learning in the network.
2) The implementation of BiPDC tends to emphasize high-frequency features, whereas standard binary convolutions generally preserve the original frequency components from the input, which are predominantly low-frequency. Emphasizing high-frequency features is more appropriate for a pyramid architecture, which processes high-level semantic information at various scales for object detection, rather than for feature extraction in the backbone.

K. Extension of Binarized Methods on Lightweight RSOD

Recent research has expanded to include the development of lightweight models for RSOD. We evaluated the performance and computational complexity of popular remote sensing object-detection methods using the AI-TOD dataset. In Fig. 8, we examined recently proposed methods such as KLDet [90], MEDet [91], and FFCA-YOLO [92]. KLDet introduces the Kullback–Leibler divergence, which effectively captures minor deviations in small bounding boxes and measures non-overlapping areas, thus enhancing the detection of tiny objects by identifying more potential positive samples. MEDet employs a center-region-based label assignment strategy that eliminates the need for a region proposal network. This method considers anchors within the central region of ground truth boxes as positive samples, increasing the detection rate of small objects, and incorporates a gated context aggregation module to selectively enhance feature representation by integrating valuable contextual information. FFCA-YOLO [92] features three innovative lightweight modules designed to enhance local region perception, multiscale feature fusion, and global association across channels and spatial dimensions, all while maintaining minimal complexity. This approach improves the detection of small objects and reduces background confusion. Further, in Fig. 9, we explored the performance of FFCA-YOLO on the AI-TOD dataset, along with its binarized version, implementing the newly proposed HBiPiDiNet on FFCA-YOLO. The experiments confirm the effectiveness of our method across popular lightweight remote sensing object-detection networks, suggesting broad applicability and ease of implementation. We also discuss the benefits of the binarized network and justify the relevance of this study. Owing to variations in experimental frameworks, we standardized experimental conditions across all methods by using a uniform anchor box allocation and excluding the use of RFLA.

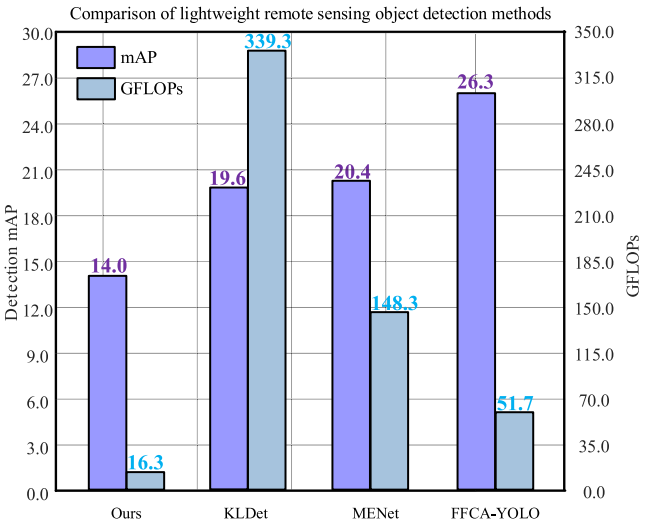


Fig. 8. Comparison of mAP and GFLOPs for popular RSOD models.

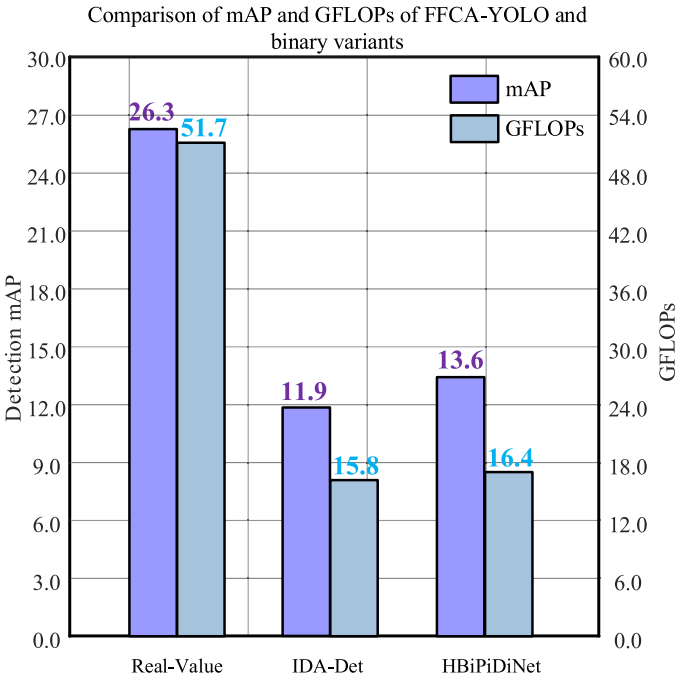


Fig. 9. Comparison of mAP and GFLOPs for three FFCA-YOLO variants: real-valued counterpart, IDa-Det applied, and HBiPiDiNet applied.

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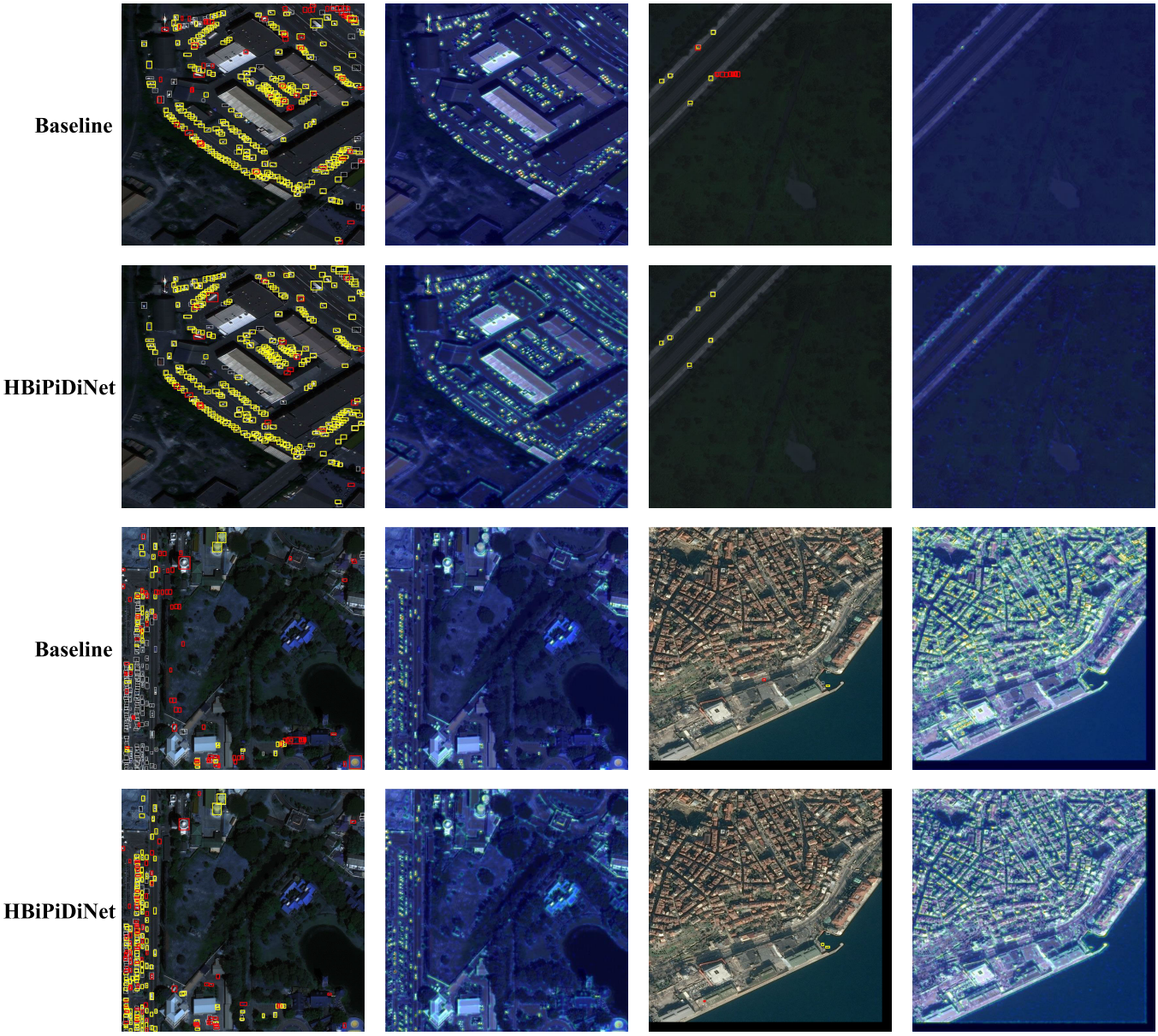


Fig. 10. Detection results and heatmap of AI-TOD. For each scene, the red, white, and yellow boxes represent the false, miss, and right detections, respectively. On the heatmap, yellow-highlighted areas represent the regions of interest, with higher brightness indicating stronger responses. The enhanced local feature response significantly improves the ability of the model to detect small objects, demonstrating the suitability of our method for remote sensing scenarios.

L. Visualization

Binarized networks often exhibit deficiencies in their feature representation. The proposed method aims to enhance the feature response of binarized networks in RSOD scenarios and improve their ability to perceive objects that are unevenly distributed. To illustrate the benefits of HBiPiDiNet in RSOD and confirm its suitability for remote sensing contexts, we present visual experimental results using the AI-TOD dataset as an example. Additionally, we utilize heatmaps to demonstrate how HBiPiDiNet integrates the LBP operator with convolution to leverage local contrast priors, thereby enhancing key local responses and improving the detection of unevenly distributed objects. This adaptation renders the model more effective for remote sensing applications, enhancing its performance and reducing both FPs and negatives. This analysis facilitates a comparison of how our proposed method and the baseline model perceive key image areas, offering an intuitive understanding of the advantages of the proposed method in RSOD. Fig. 10 displays the prediction results on the AI-TOD dataset with both the score threshold and IoU threshold set at 0.5. The detection results reveal that: 1) HBiPiDiNet achieves significantly stronger feature responses than the baseline by focusing on key local areas, even when the objects are very small, thereby confirming the effectiveness of BiPDC's emphasis on contrasting prior information. 2) In areas of the heatmaps where tiny objects exhibited weak responses nearly indistinguishable from the background, HBiPiDiNet activated these responses, demonstrating HKFCB's effectiveness in enhancing scale prior information. 3) HBiPiDiNet notably reduced FPs and FNs, as evidenced by the enhanced local area responses

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TABLE VII

COMPARE THE TIME CONSUMPTION ON HARDWARE FOR THE REAL-VALUED NETWORK, IDA-DET, AND HBIPIDINET. TIME COST REPRESENTS THE AVERAGE INFERENCE TIME PER IMAGE, CALCULATED BY INFERRING 2000 IMAGES AND TAKING THE MEAN

Framework	Backbone	Method	W/A	Time consume (s)	Acceleration
Faster-RCNN	ResNet-18	Real-valued	32/32	43.10	-
		IDa-Det	1/1	9.62	4.48×
		HBiPiDiNet	1/1	11.16	3.86×
	ConvNext-T	Real-valued	32/32	59.47	-
		IDa-Det	1/1	20.20	2.94×
		HBiPiDiNet	1/1	23.31	2.55×
RetinaNet	ResNet-18	Real-valued	32/32	42.88	-
		IDa-Det	1/1	10.95	3.92×
		HBiPiDiNet	1/1	12.17	3.52×

shown in the heatmaps. Particularly in locations undetected by the baseline but successfully identified by HBiPiDiNet, the heatmaps display a significant improvement over the baseline.

M. Deployment Efficiency

As shown in Table VII, we implemented the 1-bit networks achieved by IDa-Det and HBiPiDiNet on Raspberry Pi 4, which includes four ARM Cortex-A72 cores with a clock speed of up to 1.5 GHz. By evaluating their real-time speeds, it has been demonstrated that binarized networks can significantly enhance efficiency on mobile devices. Our proposed HBiPiDiNet achieved a balance between accuracy and efficiency, proving our method's feasibility in achieving precision in RSOD while only causing minimal efficiency loss. In the experiments, we used the AI-TOD dataset, trained on the trainval dataset with 1 NVIDIA 3090 GPU, and tested on the test dataset on Raspberry Pi 4 using an image resolution of 800×800. The results indicate that HBiPiDiNet, implemented with the Faster RCNN framework and ResNet-18 backbone, achieved a $3.86\times$ speed increase, slightly lower than the $4.48\times$ acceleration of IDa-Det. The experimental results are close to those referenced in literature [20]. However, combined with previous experimental results, the smaller sacrifice in efficiency led to a significant improvement in accuracy, proving the effectiveness of our method.

VI. CONCLUSION

In this study, we introduce the HBiPiDiNet, a novel method designed for object detection in remote sensing images. HBiPiDiNet is tailored to exploit the unique characteristics of these images, enhancing feature responses in critical local areas and capturing more detailed object-scale information. By incorporating the handcrafted LBP operator into the BiConv, we developed a new structure termed BiPDC, which combines the properties of both components. This integration enhances the discriminative capability of the convolutions by emphasizing responses in crucial local areas. Additionally, we propose a heterogeneous kernel fusion block that broadens the receptive field of the convolutions in high-resolution stages, improving the ability of the model to perceive scale and adapt to sparsely distributed objects. This study marks the first introduction of a binary object detector specifically suited for remote sensing contexts. Extensive testing shows that HBiPiDiNet achieves superior performance compared to existing binary object detectors on competitive remote sensing benchmarks. Our comprehensive analysis confirms the effectiveness and significance of the proposed model, highlighting the benefits of merging the classic LBP operator with convolutional operators. Future research could focus on integrating LBP with emerging visual architectures, such as ViT [57] layers and Mamba [93], [94] layers, to further enhance the detection performance. Additionally, the potential for binary acceleration in foundational remote sensing models warrants further exploration.

REFERENCES

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] L. Liu et al., "Deep learning for generic object detection: A survey," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 261–318, 2020.

[3] L. Deng, G. Li, S. Han, L. Shi, and Y. Xie, "Model compression and hardware acceleration for neural networks: A comprehensive survey," *Proc. IEEE*, vol. 108, no. 4, pp. 485–532, Apr. 2020.

[4] Z. Li et al., "Deep learning-based object detection techniques for remote sensing images: A survey," *Remote Sens.*, vol. 14, no. 10, p. 2385, May 2022.

[5] G. Cheng et al., "Towards large-scale small object detection: Survey and benchmarks," 2022, *arXiv:2207.14096*.

[6] T. Choudhary, V. Mishra, A. Goswami, and J. Sarangapani, "A comprehensive survey on model compression and acceleration," *Artif. Intell. Rev.*, vol. 53, pp. 5113–5155, Feb. 2020.

[7] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," *ACM Comput. Surveys*, vol. 55, no. 6, pp. 1–28, 2022.

[8] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[9] D. Wang et al., "Advancing plain vision transformer toward remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5607315.

[10] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[11] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[12] A. Bouguettaya, A. Kechida, and A. M. Taberkit, "A survey on lightweight CNN-based object detection algorithms for platforms with limited computational resources," *Int. J. Inform. Appl. Math.*, vol. 2, no. 2, pp. 28–44, 2019.

[13] J. Pang, C. Li, J. Shi, Z. Xu, and H. Feng, "$\mathcal{R}^2$-CNN: Fast tiny object detection in large-scale remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5512–5524, Aug. 2019.

[14] R. Zhang, X. Jiang, J. An, and T. Cui, "Data-free low-bit quantization for remote sensing object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[15] C. Li, G. Cheng, G. Wang, P. Zhou, and J. Han, "Instance-aware distillation for efficient object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602011.

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[16] L. Huyan et al., "Remote sensing imagery object detection model compression via tucker decomposition," *Mathematics*, vol. 11, no. 4, p. 856, Feb. 2023.

[17] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," 2021, *arXiv:2103.13630*.

[18] C. Yuan and S. S. Agaian, "A comprehensive review of binary neural network," *Artif. Intell. Rev.*, vol. 56, no. 11, pp. 12949–13013, Nov. 2023.

[19] Z. Wang, Z. Wu, J. Lu, and J. Zhou, "BiDet: An efficient binarized object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2049–2058.

[20] S. Xu et al., "IDa-Det: An information discrepancy-aware distillation for 1-bit detectors," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 346–361.

[21] Z. Su et al., "Lightweight pixel difference networks for efficient visual representation learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14956–14974, Dec. 2023.

[22] Z. Yu et al., "Searching central difference convolutional networks for face anti-spoofing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 5295–5305.

[23] Z. Su et al., "Pixel difference networks for efficient edge detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5097–5107.

[24] J. Liu, C. Zhou, P. Chen, and C. Kang, "An efficient contrast enhancement method for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1715–1719, Oct. 2017.

[25] J. H. Jang, Y. Bae, and J. B. Ra, "Contrast-enhanced fusion of multisensor images using subband-decomposed multiscale retinex," *IEEE Trans. Image Process.*, vol. 21, no. 8, pp. 3479–3490, Aug. 2012.

[26] B. Fulkerson, A. Vedaldi, and S. Soatto, "Class segmentation and object localization with superpixel neighborhoods," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 670–677.

[27] J. Trefnỳ and J. Matas, "Extended set of local binary patterns for rapid object detection," in *Proc. Comput. Vis. Winter Workshop*, 2010, pp. 1–7.

[28] Y. He and L. Xiao, "Structured pruning for deep convolutional neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 2900–2919, May 2024.

[29] H. Cheng, M. Zhang, and J. Q. Shi, "A survey on deep neural network pruning: Taxonomy, comparison, analysis, and recommendations," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 10558–10578, Dec. 2024.

[30] M. Lin et al., "Network pruning using adaptive exemplar filters," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7357–7366, Dec. 2022.

[31] Y. Ma et al., "OMPQ: Orthogonal mixed precision quantization," in *Proc. AAAI*, Jun. 2023, vol. 37, no. 7, pp. 9029–9037.

[32] S. Li et al., "Distilling a powerful student model via online knowledge distillation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8743–8752, Mar. 2022.

[33] X. Zhang, J. Du, Y. Li, W. Xie, and J. T. Zhou, "Spanning training progress: Temporal dual-depth scoring (TDDS) for enhanced dataset pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 26213–26222.

[34] N. Kieu, K. Nguyen, A. Nazib, T. Fernando, C. Fookes, and S. Sridharan, "Multimodal colearning meets remote sensing: Taxonomy, state of the art, and future works," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7386–7409, 2024.

[35] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.

[36] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," *Remote Sens. Environ.*, vol. 299, Dec. 2023, Art. no. 113856.

[37] G. Chen et al., "A survey of the four pillars for small object detection: Multiscale representation, contextual information, super-resolution, and region proposal," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 2, pp. 936–953, Feb. 2022.

[38] W. Han et al., "Methods for small, weak object detection in optical high-resolution remote sensing images: A survey of advances and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 8–34, Dec. 2021.

[39] J. Wu, C. Zhou, Q. Zhang, M. Yang, and J. Yuan, "Self-mimic learning for small-scale pedestrian detection," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 2012–2020.

[40] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.

[41] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Cho, "Augmentation for small object detection," 2019, *arXiv:1902.07296*.

[42] C. Chen et al., "RRNet: A hybrid detector for object detection in drone-captured images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 100–108.

[43] C. Guo, B. Fan, Q. Zhang, S. Xiang, and C. Pan, "AugFPN: Improving multi-scale feature learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12595–12604.

[44] X. Zhang, W. Yang, J. Feng, B. Dai, T. Bu, and X. Lu, "GSpect: Spectral filtering for cross-scale graph classification," 2024, *arXiv:2409.00338*.

[45] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.

[46] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5604610.

[47] Y. Yang et al., "Adaptive knowledge distillation for lightweight remote sensing object detectors optimizing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623715.

[48] Y. Zhang, Z. Yan, X. Sun, W. Diao, K. Fu, and L. Wang, "Learning efficient and accurate detectors with dynamic knowledge distillation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613819.

[49] S. Chen, R. Zhan, W. Wang, and J. Zhang, "Learning slimming SAR ship object detector through network pruning and knowledge distillation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1267–1282, 2021.

[50] Y. Shen, D. Liu, F. Zhang, and Q. Zhang, "Fast and accurate multi-class geospatial object detection with large-size remote sensing imagery using CNN and truncated NMS," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 235–249, Sep. 2022.

[51] D. Soudry, I. Hubara, and R. Meir, "Expectation backpropagation: Parameter-free training of multilayer neural networks with continuous or discrete weights," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 27, Dec. 2014, pp. 963–971.

[52] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to $+1$ or -1," 2016, *arXiv:1602.02830*.

[53] M. Rastegari, V. Ordonez, J. Redmon, and A. Farhadi, "XNOR-Net: ImageNet classification using binary convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 525–542.

[54] D. H. Le and T. V. Pham, "Improving bi-real net with block-wise quantization and multiple-steps binarization on activation," in *Proc. RIVF Int. Conf. Comput. Commun. Technol. (RIVF)*, Oct. 2020, pp. 1–6.

[55] Z. Liu, Z. Shen, M. Savvides, and K.-T. Cheng, "ReActNet: Towards precise binary neural network with generalized activation functions," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 143–159.

[56] C. Lee, H. Kim, E. Park, and J.-J. Kim, "INSTA-BNN: Binary neural network with INSTAnce-aware threshold," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17325–17334.

[57] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[58] Z. Liu et al., "BiT: Robustly binarized multi-distilled transformer," 2022, *arXiv:2205.13016*.

[59] Y. Li, S. Xu, B. Zhang, X. Cao, P. Gao, and G. Guo, "Q-ViT: Accurate and fully quantized low-bit vision transformer," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 35, Jan. 2022, pp. 34451–34463.

[60] P.-H.-C. Le and X. Li, "BinaryViT: Pushing binary vision transformers towards convolutional models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 4664–4673.

[61] Y. He et al., "BiViT: Extremely compressed binary vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 5651–5663.

[62] S. Xu, Z. Liu, X. Gong, C. Liu, M. Mao, and B. Zhang, "Amplitude suppression and direction activation in networks for 1-bit faster R-CNN," in *Proc. 4th Int. Workshop Embedded Mobile Deep Learn.*, Sep. 2020, pp. 19–24.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 13:41:23 UTC from IEEE Xplore. Restrictions apply.

[63] T. Ojala, M. Pietikainen, and T. Maenpaa, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, Jul. 2002.

[64] F. Juefei-Xu, V. N. Boddeti, and M. Savvides, "Local binary convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 19–28.

[65] L. Liu, L. Zhao, Y. Long, G. Kuang, and P. Fieguth, "Extended local binary patterns for texture classification," *Image Vis. Comput.*, vol. 30, no. 2, pp. 86–99, Feb. 2012.

[66] J. Zhang, Z. Su, and L. Liu, "Median pixel difference convolutional network for robust face recognition," 2022, *arXiv:2205.15867*.

[67] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 526–543.

[68] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-net in U-net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[69] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2019.

[70] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31: Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11963–11975.

[71] S. Liu et al., "More ConvNets in the 2020s: Scaling up kernels beyond 51×51 using sparsity," 2022, *arXiv:2207.03620*.

[72] S. Gao, Z.-Y. Li, Q. Han, M.-M. Cheng, and L. Wang, "RF-Next: Efficient receptive field search for convolutional neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 2984–3002, Mar. 2023.

[73] Y. Chen, X. Yuan, R. Wu, J. Wang, Q. Hou, and M.-M. Cheng, "YOLO-MS: Rethinking multi-scale representation learning for real-time object detection," 2023, *arXiv:2308.05480*.

[74] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[75] D. Du et al., "VisDrone-DET2019: The vision meets drone object detection in image challenge results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 213–226.

[76] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[77] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.

[78] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[79] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.

[80] W. Yu et al., "MetaFormer is actually what you need for vision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10819–10829.

[81] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[82] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 11976–11986.

[83] Z. Liu, W. Luo, B. Wu, X. Yang, W. Liu, and K.-T. Cheng, "Bi-Real net: Binarizing deep network towards real-network performance," *Int. J. Comput. Vis.*, vol. 128, no. 1, pp. 202–219, Jan. 2020.

[84] S. K. Esser, J. L. McKinstry, D. Bablani, R. Appuswamy, and D. S. Modha, "Learned step size quantization," 2019, *arXiv:1902.08153*.

[85] Z. Tu, X. Chen, P. Ren, and Y. Wang, "AdaBin: Improving binary neural networks with adaptive binary sets," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2022, pp. 379–395.

[86] S. Xu et al., "Resilient binary neural network," in *Proc. AAAI*, Jun. 2023, vol. 37, no. 9, pp. 10620–10628.

[87] Z. Xu et al., "ReCU: Reviving the dead weights in binary neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 5198–5208.

[88] D. Jha et al., "IRNet: A general purpose deep residual regression framework for materials discovery," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Jul. 2019, pp. 2385–2393.

[89] H. Xu et al., "Senputing: An ultra-low-power always-on vision perception chip featuring the deep fusion of sensing and computing," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 1, pp. 232–243, Jan. 2022.

[90] Z. Zhou and Y. Zhu, "KLDet: Detecting tiny objects in remote sensing images via Kullback–Leibler divergence," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703316.

[91] T. Zhang et al., "Multistage enhancement network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611512.

[92] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[93] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[94] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.



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